

No transport may exist The first one is very easy. Consider the case $\mu = \delta_a$, $a \in X$ and suppose that ν is not a Dirac mass. In this case there is no transport map at all. Indeed, it is easy to check that $T_{\#}\delta_a = \delta_{T(a)}$ and hence no transport map $T : X \rightarrow Y$ can exist if ν is not of the form δ_b for some $b \in Y$.

More generally, we can say that the image measure $T_{\#}\mu$ always includes an atom of mass at least $\mu(\{a\})$ for every atom a of μ . This implies in particular that measures with atoms cannot be sent through a transport map onto measures without atoms. For these reasons, the absence of atoms is a typical assumption on the source measure μ when one wants to solve (MP).

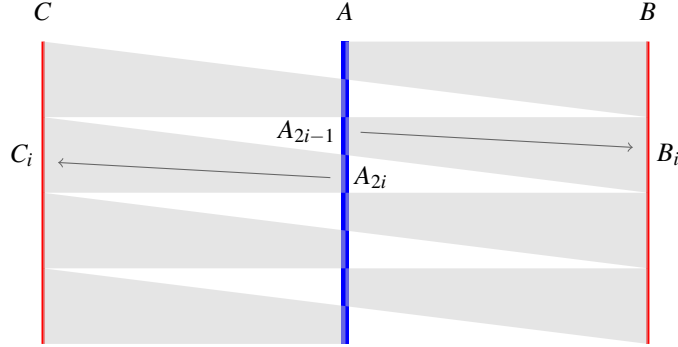
On the contrary, the problem (KP) by Kantorovich still makes sense even for atomic measures. In particular, the situation is very easy when $\mu = \delta_a$. In this case the set $\Pi(\mu, \nu)$ contains a unique element, which is $\gamma = \mu \otimes \nu = \delta_a \otimes \nu$.

Transport can exist, but no optimal one We consider the quadratic cost $c(x, y) = |x - y|^2$ (but any other cost strictly increasing in the distance would give the same result). Set

$$\mu = \mathcal{H}^1 \llcorner A \text{ and } \nu = \frac{\mathcal{H}^1 \llcorner B + \mathcal{H}^1 \llcorner C}{2}$$

where A , B and C are three vertical parallel segments in $\mathbb{R}^2$ whose vertexes lie on the two lines $y = 0$ and $y = 1$ and the abscissas are 0, 1 and -1 , respectively, and $\mathcal{H}^1$ is the 1-dimensional Haudorff measure. It is clear that no transport plan may realize a cost better than 1 since, horizontally, every point needs to be displaced of a distance 1. Moreover, one can get a sequence of maps $T_n : A \rightarrow B \cup C$ by dividing A into $2n$ equal segments $(A_i)_{i=1, \dots, 2n}$ and B and C into n segments each, $(B_i)_{i=1, \dots, n}$ and $(C_i)_{i=1, \dots, n}$ (all ordered downwards). Then define T_n as a piecewise affine map which sends A_{2i-1} onto B_i and A_{2i} onto C_i . In this way the cost of the map T_n is less than $1 + 1/n$, which implies that the infimum of the Kantorovich problem is 1, as well as the infimum on transport maps only. Yet, no map T may obtain a cost 1, as this would imply that all points are sent horizontally, but this cannot respect the push-forward constraint. On the other hand, the transport plans associated to T_n weakly converge to the transport plan $\frac{1}{2}\gamma_T^+ + \frac{1}{2}\gamma_T^-$ where $T^\pm(x) = x \pm e$ and $e = (1, 0)$. This transport plan turns out to be the only optimal transport plan and its cost is 1.

Note that in this last example we also saw that the infimum among transport maps was equal to the infimum (i.e. the minimum) among transport plans. This is a general fact, which relies on the notion of relaxation, as we can see in the following section.



1.5 Kantorovich as a relaxation of Monge

Let us set again $K(\gamma) := \int_{\Omega \times \Omega} c \, d\gamma$. Since we know that for any map T we have $M(T) = \int_{\Omega} c(x, T(x)) \, d\mu = \int_{\Omega \times \Omega} c \, d\gamma_T = K(\gamma_T)$, the Monge Problem may be re-written as

$$\min \{J(\gamma) : \gamma \in \Pi(\mu, \nu)\},$$

where

$$J(\gamma) = \begin{cases} K(\gamma) = M(T) & \text{if } \gamma = \gamma_T, \\ +\infty & \text{otherwise.} \end{cases}$$

This is simple to understand: the definition of J forces to restrict the minimization to those plan induced by a transport map. This fact is useful in order to consider (MP) and (KP) as two problems on the *same set of admissible objects*, where the only difference is the functional to be minimized, J or K .

The question is now : why did Kantorovich decide to replace J with K ? Can we easily prove that $\inf K = \inf J$? This is obviously true when, by chance, the minimizer of K is of the form $\gamma = \gamma_T$, since in this case we would have equality of the two minima. But is it possible to justify the procedure in general? The main mathematical justification comes from the following notion of *relaxation*.

Box 1.10. – Memo – Relaxation

Let $F : X \rightarrow \mathbb{R} \cup \{+\infty\}$ be a given functional on a metric space X , and suppose that it is bounded from below. We define the relaxation of F as the functional $\bar{F} : X \rightarrow \mathbb{R} \cup \{+\infty\}$ which is the maximal functional among those $G : X \rightarrow \mathbb{R} \cup \{+\infty\}$ which are lower semi-continuous and such that $G \leq F$. This functional exists since the supremum of an arbitrary family of l.s.c. functions is also l.s.c. Moreover, we also have a representation formula, which is easy to prove:

$$\bar{F}(x) = \inf \{ \liminf_n F(x_n) : x_n \rightarrow x \}.$$

A consequence of the definition is also that $\inf F = \inf \bar{F}$ (this latter infimum, that of $\bar{F}$, being often a minimum, when X is compact). This is easy to check: $F \geq \bar{F}$ implies $\inf F \geq$

$\inf \bar{F}$; on the other hand we have $F \geq \ell$ where $\ell := \inf F$, and a constant function is l.s.c. Hence $\bar{F} \geq \ell$ and $\inf \bar{F} \geq \inf F$.

Here we claim that, under some assumptions, K is actually the relaxation of J . It will happen, in this case, by chance, that this relaxation is also continuous, instead of only semi-continuous, and that it coincides with J on $\{J < +\infty\}$.

The assumptions are the following: we take $\Omega \subset \mathbb{R}^d$ to be compact, c continuous and μ atomless (i.e. for every $x \in \Omega$ we have $\mu(\{x\}) = 0$). Compactness is supposed for simplicity, while the result could be true under more general assumptions. The reference on this subject are [253] and [8].

We need some preliminary results. The first concerns the one-dimensional case, that we will analyze in details in Chapter 2, but can be stated right now.

Lemma 1.27. *If μ, ν are two probability measures on the real line $\mathbb{R}$ and μ is atomless, then there exists at least a transport map T such that $T_{\#}\mu = \nu$.*

Proof. Just consider the monotone increasing map T provided by Remark 1.23. This map also optimizes the quadratic cost, but here we don't care about it. $\square$ $\square$

Lemma 1.28. *There exists a Borel map $\sigma_d : \mathbb{R}^d \rightarrow \mathbb{R}$ which is injective, its image is a Borel subset of $\mathbb{R}$, and its inverse map is Borel measurable as well.*

Proof. First note that it is sufficient to prove this result for $d = 2$, since then one can proceed by induction: if a map σ_{d-1} is given on $\mathbb{R}^{d-1}$, then one can produce a map σ_d by considering $\sigma_d(x_1, x_2, \dots, x_d) = \sigma_2(x_1, \sigma_{d-1}(x_2, x_3, \dots, x_d))$.

Then, note also that it is enough to define such a map on $]0, 1[^2$, since one can go from $\mathbb{R}^2$ to $]0, 1[^2$ by considering $(x, y) \mapsto (\frac{1}{2} + \frac{1}{\pi} \arctan x, \frac{1}{2} + \frac{1}{\pi} \arctan y)$.

Then, consider the map which associates to the pair (x, y) , where $x = 0.x_1x_2x_3\dots$ and $y = 0.y_1y_2y_3\dots$ in decimal (or binary) notation, the point $0.x_1y_1x_2y_2x_3y_3\dots$. In order to avoid ambiguities, we can decide that no decimal notation is allowed to end with a periodic 9 (i.e. $0.34729999\dots$ has to be written as 0.3473). This is why the image of this map will not be the whole interval, since the points like $0.39393939\dots$ are not obtained through this map. But this set of points is actually Borel measurable.

It is not difficult neither to check that the map is Borel measurable, as well as its inverse, since the pre-image of every interval defined by prescribing the first $2k$ digits of a number in $\mathbb{R}$ is just a rectangle in $\mathbb{R}^2$, the product of two intervals defined by prescribing the first k digits of every component. These particular intervals being a base for the Borel tribe, this proves the measurability we need. $\square$ $\square$

Corollary 1.29. *If μ, ν are two probability measures on $\mathbb{R}^d$ and μ is atomless, then there exists at least a transport map T such that $T_{\#}\mu = \nu$.*

Proof. This is just obtained by considering a transport map T from $(\sigma_d)_{\#}\mu$ to $(\sigma_d)_{\#}\nu$ and then composing with σ_d and $(\sigma_d)^{-1}$. $\square$ $\square$

Remark 1.30. We provided here, through Lemma 1.28, an explicit way of constructing a transport map between an atomless measure μ and an arbitrary measure ν , when the ambient space is $\mathbb{R}^d$. Actually, this fact is much more general, since it is well known that any measure space endowed with an atomless measure is isomorphic to $[0, 1]$ with the Lebesgue measure⁴. Yet, we do not want to introduce this kind of arguments when an explicit construction is sufficient to deal with the Euclidean case, which already meets our scopes.

A last lemma:

Lemma 1.31. *Consider on a compact metric space X , endowed with a probability $\rho \in \mathcal{P}(X)$, a sequence of partitions G_n , each G_n being a family of disjoint subsets $C_{i,n}$ such that $\bigcup_{i \in I_n} C_{i,n} = X$ for every n . Suppose that $\text{size}(G_n) := \max_i \text{diam}(C_{i,n})$ tends to 0 as $n \rightarrow \infty$ and consider a sequence of probability measures ρ_n on X such that, for every n and $i \in I_n$, we have $\rho_n(C_{i,n}) = \rho(C_{i,n})$. Then $\rho_n \rightarrow \rho$.*

Proof. Set $m_{i,n} := \rho_n(C_{i,n}) = \rho(C_{i,n})$. It is sufficient to take a continuous function $\phi \in C(X)$ and note that

$$\begin{aligned} \left| \int_X \phi d\rho_n - \int_X \phi d\rho \right| &\leq \sum_{i \in I_n} \left| \int_{C_{i,n}} \phi d\rho_n - \int_{C_{i,n}} \phi d\rho \right| \\ &\leq \omega(\text{diam}(C_{i,n})) \sum_{i \in I_n} m_{i,n} = \omega(\text{diam}(C_{i,n})) \rightarrow 0, \end{aligned}$$

where ω is the modulus of continuity of ϕ . This is justified by the fact that, whenever two measures have the same mass on a set $C \subset X$, since the oscillation of ϕ on the same set does not exceed $\omega(\text{diam}(C))$, the difference of the two integrals is no more than this number times the common mass.

This proves $\int \phi d\rho_n \rightarrow \int \phi d\rho$ and hence $\rho_n \rightarrow \rho$. □ □

We can now prove the following

Theorem 1.32. *On a compact subset $\Omega \subset \mathbb{R}^d$, the set of plans γ_T induced by a transport is dense in the set of plans $\Pi(\mu, \nu)$ whenever μ is atomless.*

Proof. Fix n , and consider any partition of Ω into sets $K_{i,n}$ of diameter smaller than $1/(2n)$ (for instance, small cubes). The sets $C_{i,j,n} := K_{i,n} \times K_{j,n}$ make a partition of $\Omega \times \Omega$ with size smaller than $1/n$.

Let us now take any measure $\gamma \in \Pi(\mu, \nu)$. Thanks to Lemma 1.31, we only need to build a transport T sending μ to ν , such that γ_T gives the same mass as γ to each one of the sets $C_{i,j,n}$. To do this, define the columns $Col_{i,n} := K_{i,n} \times \Omega$ and denote by $\gamma_{i,n}$ the restriction of γ on $Col_{i,n}$. Its marginal will be denoted by $\mu_{i,n}$ and $\nu_{i,n}$. Consider now, for each i , a transport map $T_{i,n}$ sending $\mu_{i,n}$ to $\nu_{i,n}$. It exists thanks to Corollary 1.29, since for each (i, n) we have $\mu_{i,n} \leq \mu$, which makes these measures atomless. Since the

⁴Note that this was the argument used in [176] to obtain the existence of transport maps in general spaces, and [253] points out that, when composing with the isomorphisms sending arbitrary spaces to $[0, 1]$, the cost function is no more continuous. Yet, this could be handled using Lemma 1.8, which allows to use costs of this form: the composition of a continuous function with measurable maps of x and y , separately.

$\mu_{i,n}$ are concentrated on disjoint sets, by “gluing” the transports $T_{i,n}$ we get a transport T sending μ to ν (using $\sum_i \mu_{i,n} = \mu$ and $\sum_i \nu_{i,n} = \nu$).

It is enough to check that γ_T gives the same mass⁵ as γ to every $C_{i,j,n}$, but it is easy to prove. Indeed, this mass equals that of $\gamma_{T_{i,n}}$ and $\gamma_{T_{i,n}}(C_{i,j,n}) = \mu_{i,n}(\{x : x \in K_{i,n}, T_{i,n}(x) \in K_{j,n}\}) = \mu_{i,n}(\{x : T_{i,n}(x) \in K_{j,n}\}) = \nu_{i,n}(K_{j,n}) = \gamma(K_{i,n} \times K_{j,n})$. $\square$ $\square$

The relaxation result is just a consequence.

Theorem 1.33. *Under the above-mentioned assumptions, K is the relaxation of J . In particular, $\inf J = \min K$, and hence Monge and Kantorovich problems have the same infimum.*

Proof. First note that, since K is continuous, then it is l.s.c. and since we have $K \leq J$, then K is necessarily smaller than the relaxation of J . We only need to prove that, for each γ , we can find a sequence of transports T_n such that $\gamma_{T_n} \rightarrow \gamma$ and $J(\gamma_{T_n}) \rightarrow K(\gamma)$, so that the infimum in the sequential characterization of the relaxed functional (see definition) will be smaller than K , thus proving the equality. Actually, since for $\gamma = \gamma_{T_n}$ the two functionals K and J coincide, and since K is continuous, we only need to produce a sequence T_n such that $\gamma_{T_n} \rightarrow \gamma$. This is possible thanks to Theorem 1.32 $\square$ $\square$

1.6 Convexity, c -concavity, cyclical monotonicity, duality and optimality

1.6.1 Convex and c -concave functions

In this section we analyze properties of c -concave functions in comparison with convex functions, which are better known. We start from recalling some notions from convex analysis.

Box 1.11. – Memo – Convex functions

The definition of convex function is always the same: $f : \mathbb{R}^d \rightarrow \mathbb{R} \cup \{+\infty\}$ is convex if and only if for all $x, y \in \mathbb{R}^d$ and $t \in [0, 1]$ we have $f((1-t)x + ty) \leq (1-t)f(x) + tf(y)$. We do not care about convex functions defined on subsets $\Omega \subset \mathbb{R}^d$ (which should be convex, for the definition to make sense) since one can always extend f outside Ω by setting it to $+\infty$, preserving convexity.

An easy stability property is the following: if f_α is a family of convex functions, then f defined through $f(x) := \sup_\alpha f_\alpha(x)$ is also convex. This can be checked easily by writing $f_\alpha((1-t)x + ty) \leq (1-t)f_\alpha(x) + tf_\alpha(y) \leq (1-t)f(x) + tf(y)$ and passing to the sup in α . Otherwise, we can also use the epigraphs (and a function is convex if and only if its epigraph is convex) and note that the interesection of convex sets is still convex.

⁵Note that [8] and [253] proved the same fact in a more complicated way: indeed, the use of a unique transport map for every column is enough. The same idea has been used in [147] for a different density result, concerning Lipschitz transport maps.